Bachelor's thesis

ML & HFTS

Predicting short-term regime shifts via HMMs

Egil Furnes
egilsf@uio.no

Department of Mathematics
Faculty of Mathematics and Natural Sciences

9th February 2026

Abstract

The LaTeX document class was created to facilitate the writing of your thesis according to the graphical guidelines of the University of Oslo.

1 Introduction

Hello there Hastie et al. (2009). This is a citation (Hastie et al. 2009). This is also a reference from Hautsch (2012), who has written a book.

Hamilton (1989) did say to Hamilton (1994) that Lo and MacKinlay (1990) and Engle (1982) might know what Jacod et al. (2017) and Aït-Sahalia and Jacod (2014) and Rabiner (1989) did to Cappé et al. (2005) although maybe not (Tsay 2010).

1.1 Bid–ask bounce

1.2 Order flow imbalance

1.3 Asynchronus trading

2 Method

Let P_τ denote the observed transaction price at tick time τ. The observed price can be decomposed as

$$P_\tau = P_\tau^* + \eta_\tau$$

where P_τ is the latent efficient price and η_τ represents microstructure noise. The pre-averaged price is defined as

$$\bar{P}_t = \frac{1}{k} \sum_{j=0}^{k-1} P_{t-j}$$

where t now indexes the last tick in each averaging block. Returns are then constructed as

$$r_t = \bar{P}_t - \bar{P}_{t-1}$$

2.1 Baseline Auto-regressive Model

As a benchmark, we consider a single-regime auto-regressive model of order one

$$r_t = \mu + \phi r_t t - 1 + \varepsilon_t$$

where μ is the unconditional mean return, ϕ measures linear serial dependence and ε_t is an innovative term with $\varepsilon_t \sim \mathcal{N}(0, \sigma^2)$.

2.2 Hidden Markov Model with Regime-Switching Auto-regressive Dynamics

Let $S_t \in \{1, \ldots, K\}$ denote an unobserved discrete-time Markov chain representing the dependence regime at time t. The transition probabilities are defined as

$$P(S_t = j \mid S_{t-1} = i) = p_{ij}$$

with $p_{ij} \geq 0$ and $\sum_{j=1}^{K} p_{ij} = 1$ for all i.

Conditional on the regime $S_t = k$, returns follow an auto-regressive process

$$r_t = \mu_k + \phi_k r_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \sigma_k^2)$$

Here μ_k is the regime-specific mean, ϕ_k captures the strength of the short-term dependence in regime k and σ_k^2 is the regime-specific innovation variance.

Regimes with ϕ_k close to zero are interpreted as noise-dominated periods, while regimes with larger positive ϕ_k correspond to short-term momentum or price returns.

2.3 Modeling volatility

Allowing σ_k^2 to vary across regimes captures abrupt changes in market uncertainty. As an extension, conditional heteroskedasticity can be introduced via a regime-dependent ARCH specification

$$\sigma_{k,t}^2 = \omega_k + \alpha_k \varepsilon_{t-1}^2$$

where $\omega_k \geq 0$ and $\alpha_k \geq 0$. Building on this, volatility persistence can be incorporated through a regime-dependent GARCH specification,

$$\sigma_{k,t}^2 = \omega_k + \alpha_k \varepsilon_{t-1}^2 + \beta_k \sigma_{k,t-1}^2,$$

where $\beta_k \geq 0$. This extension allows volatility shocks to decay gradually over time within each regime rather than affecting only the next period.

2.4 Data

Data was retrieved from HistData.com (2026).

3 Result

4 Conclusion

References

Aït-Sahalia, Yacine and Jean Jacod (2014). *High-Frequency Financial Econometrics*. Princeton University Press. URL: https://press.princeton.edu/books/hardcover/9780691161433/high-frequency-financial-econometrics.

Cappé, Olivier, Eric Moulines and Tobias Rydén (2005). *Inference in Hidden Markov Models*. Springer. DOI: 10.1007/b136369. URL: https://doi.org/10.1007/b136369.

Engle, Robert F. (1982). 'Autoregressive Conditional Heteroskedasticity with Estimates of the Variance of United Kingdom Inflation'. In: *Econometrica* 50.4, pp. 987–1007. DOI: 10.2307/1912773. URL: https://doi.org/10.2307/1912773.

Hamilton, James D. (1989). 'A New Approach to the Economic Analysis of Nonstationary Time Series and the Business Cycle'. In: *Econometrica* 57.2, pp. 357–384. DOI: 10.2307/1912559. URL: https://doi.org/10.2307/1912559.

— (1994). *Time Series Analysis*. Princeton University Press. URL: https://press.princeton.edu/books/hardcover/9780691042893/time-series-analysis.

Hastie, Trevor, Robert Tibshirani and Jerome Friedman (2009). *The Elements of Statistical Learning*. 2nd ed. Springer. ISBN: 978-0387848570.

Hautsch, Nikolaus (2012). *Econometrics of Financial High-Frequency Data*. Berlin, Heidelberg: Springer. ISBN: 978-3-642-21925-2. DOI: 10.1007/978-3-642-21925-2.

HistData.com (2026). *Free Forex Historical Data*. Accessed 2026-02-09. HistData.com. URL: https://www.histdata.com/ (visited on 09/02/2026).

Jacod, Jean et al. (2017). 'Between Data Cleaning and Inference: Pre-averaging and Robust Estimators of the Efficient Price'. In: *Journal of Econometrics* 196.1, pp. 1–22. DOI: 10.1016/j.jeconom.2016.08.014. URL: https://doi.org/10.1016/j.jeconom.2016.08.014.

Lo, Andrew W. and A. Craig MacKinlay (1990). 'An Econometric Analysis of Nonsynchronous Trading'. In: *Journal of Econometrics* 45.1-2, pp. 181–211. DOI: 10.1016/0304-4076(90)90098-E. URL: https://doi.org/10.1016/0304-4076(90)90098-E.

Rabiner, Lawrence R. (1989). 'A Tutorial on Hidden Markov Models and Selected Applications in Speech Recognition'. In: *Proceedings of the IEEE* 77.2, pp. 257–286. DOI: 10.1109/5.18626. URL: https://doi.org/10.1109/5.18626.

Tsay, Ruey S. (2010). *Analysis of Financial Time Series*. 3rd ed. Wiley. URL: https://www.wiley.com/en-us/Analysis+of+Financial+Time+Series%2C+3rd+Edition-p-9780470414354.

5 Appendix

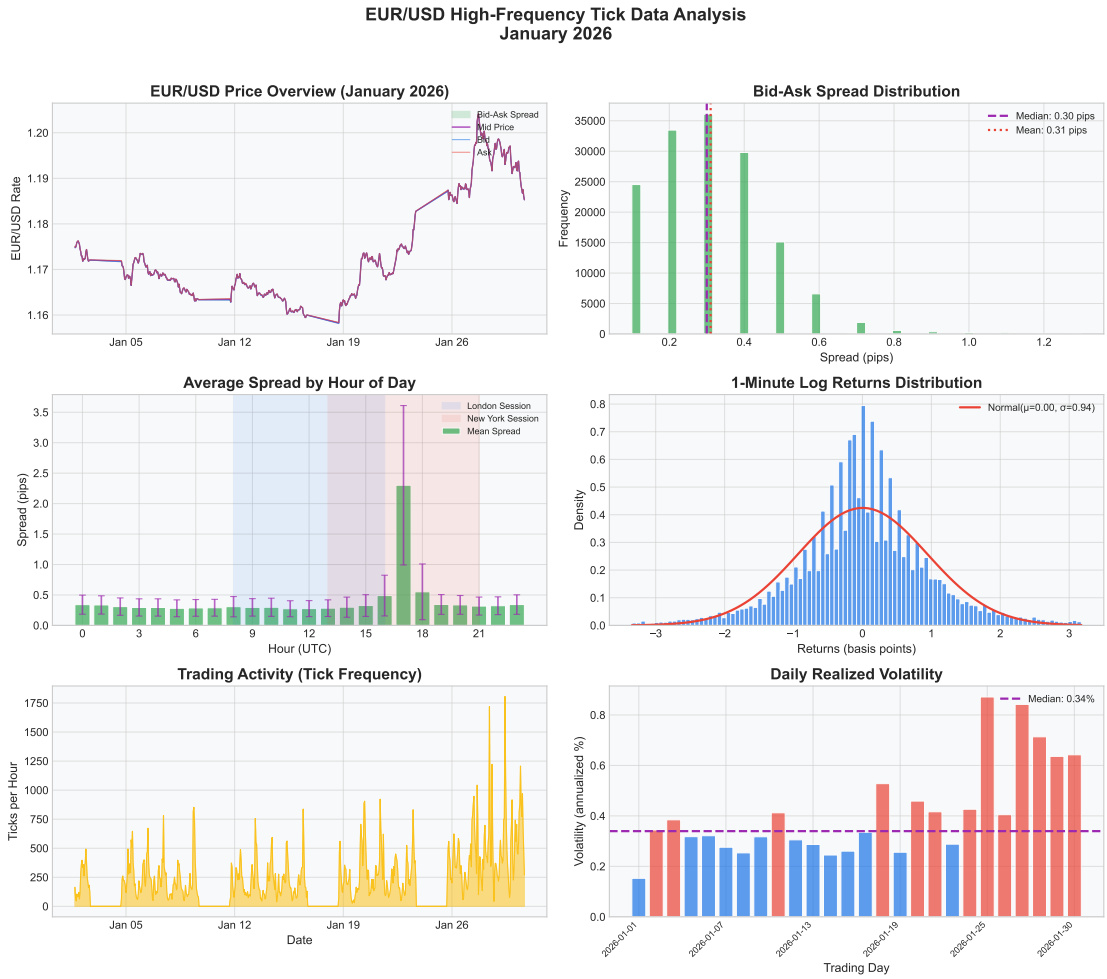

Figure 1: EURUSD Tick Analysis